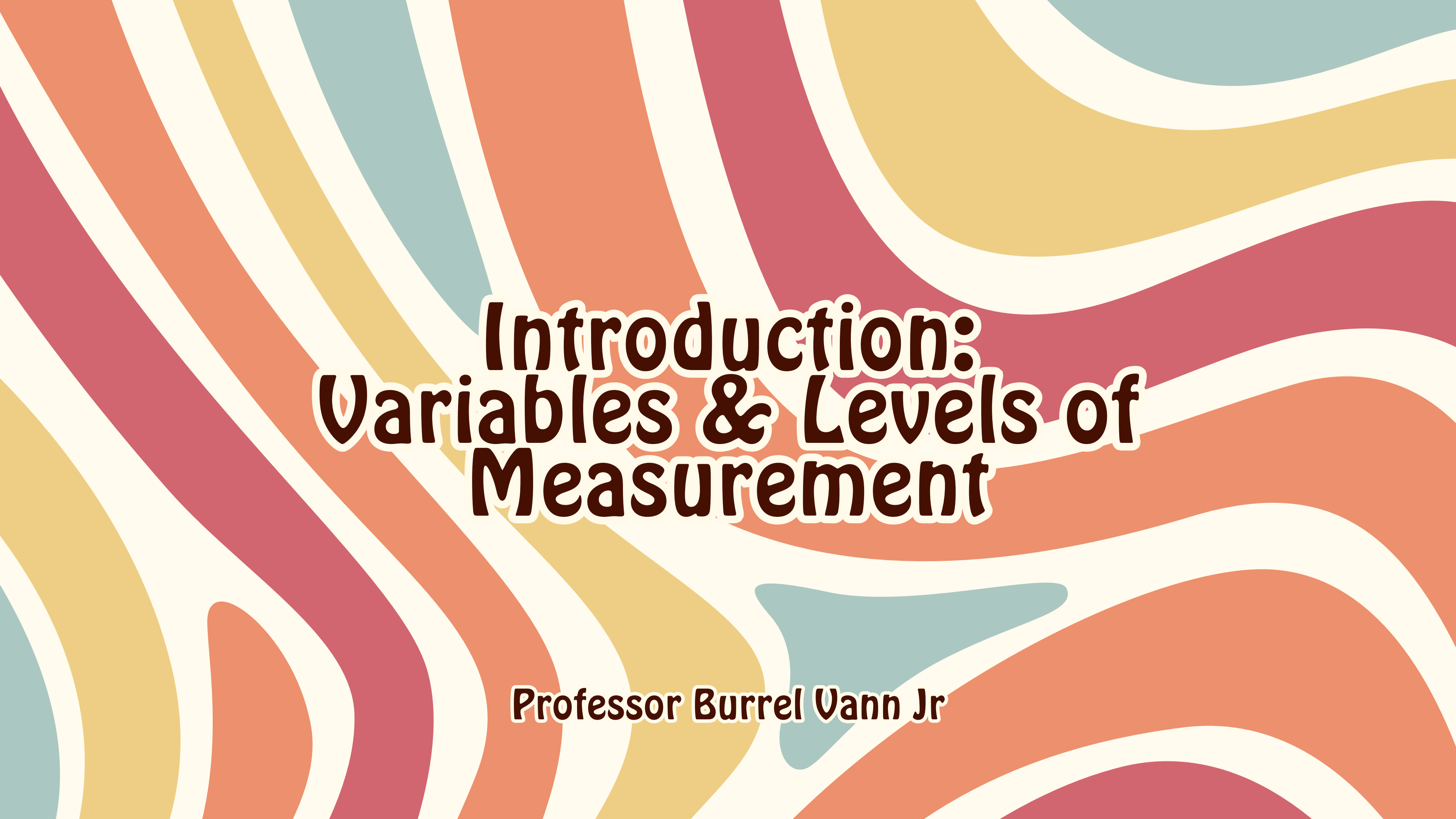




Introduction: Variables & Levels of Measurement

Professor Burrell Vann Jr

TOPICS

- Concepts & measurement
- Operationalization
- Variables: Dependent vs. independent
- Levels of measurement
- Data collection (population vs. sample)

Concepts & Measurement

In social science we try to measure abstract concepts

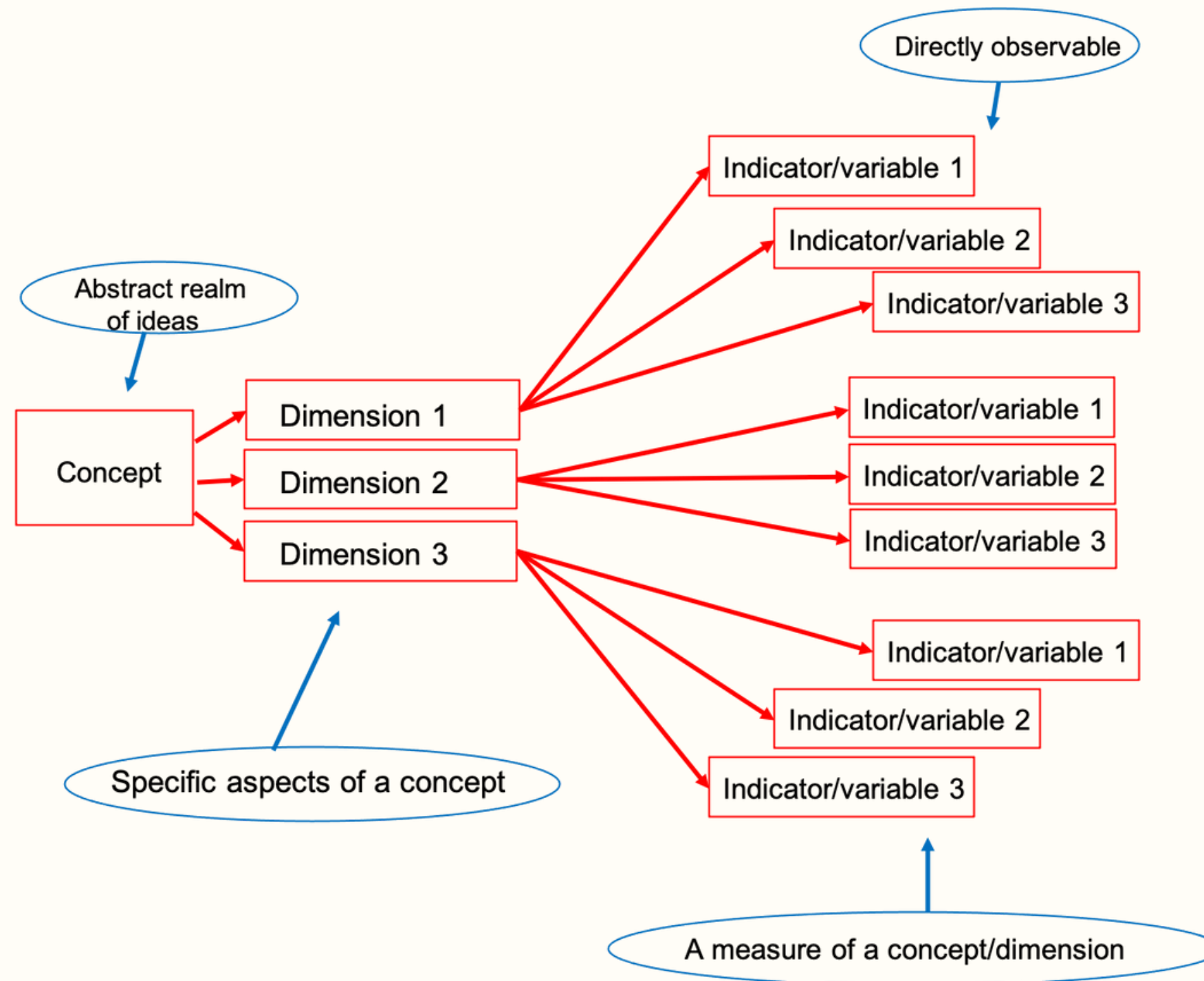
- Some examples include: Democracy, effectiveness, satisfaction, trust, prejudice, criminal subculture, occupational burnout, engagement, racism, alienation, street-crime, deviance, genocide, etc.

Concepts are constructs, developed out of theory

How can we measure abstract concepts?

- Operationalization: Process by which a researcher defines how a concept is measured and observed within a particular research project

Operationalization: From Concepts to Measures



Operationalization Example

Concept: School violence

- Definition: "Intentional use of power, taking place within educational settings, against another person, group, or community, with the behavior likely to cause physical or psychological harm."

Dimension 1: Verbal assault

- Indicators:
 - number of harassment allegations
 - number of times students called derogatory names

Note: Unit of analysis implied is the school-level. Each observation/case would be a different school

Operationalization Example: Other Dimensions

Dimension 2: Use of Weapons

- Indicators:
 - Number of students caught carrying weapons
 - Student perception of presence of weapons
 - e.g. ask students whether they perceive a low, moderate, or high presence of weapons in the educational setting

Dimension 3: Physical Contact

- Indicators:
 - Number of fights detected by administration
 - Prevalence of self-reported involvement in physical fight

Important Terms

Unit of Analysis:

- The level of observation. What does each case/observation represent (e.g. a person, an organization, a family, a school, etc.)?

Variable:

- Any empirically observable item (indicator) that can change values/attributes from case to case

Constant:

- The opposite of a variable. An observable item whose value remains unchanged

Attributes:

- The categories within a variable that specify the ways a variable can vary. The possible values cases can have for a given variable.

Instrument:

- Measurement device (survey, test, questionnaire, interview protocol etc.)

Validity:

- The notion that our instrument actually measures what it is intended to measure

Level of measurement:

- Classification of a variable that describes the level of “sophistication” of the attributes/values of that variable

Variables and Attributes

Variable	Attributes
Sex	Male, Female, Other
Age	0-125 (years old)
Age	Child, Adolescent, Young Adult, Adult
Religion	Catholic, Protestant, Evangelical
Education	Less than high school, GED/high school diploma, Some college, college graduate, some graduate school, graduate or professional degree
Education	0-30 (years of education)
Marital Status	Never Married, Married, Widowed, Divorced, Separated

Variables' Levels of Measurement

Level of Sophistication	Type	Level	Characteristics of Attributes	Examples
	Categorical	Nominal	Can only classify. Cannot be rank-ordered	Race, Sex, Religion, Political Affiliation
		Ordinal	Can <u>classify</u> and be logically <u>rank-ordered</u> but lack an associated numerical value. Distance between attributes cannot be determined.	Socioeconomic Status (SES), Police Officer Rank
	Continuous	Interval	Can <u>classify</u> , be logically <u>rank-ordered</u> , and have an associated <u>numerical value</u> , but don't have a true/meaningful zero point. Have equal numerical distance between values.	IQ scores, temperature (°F or °C)
		Ratio	Can <u>classify</u> , be logically <u>rank-ordered</u> , have an associated <u>numerical value</u> , and have a <u>true/meaningful zero point</u> . Have equal numerical distance between values.	number of children, age, pulse, years on the police force

Levels of Measurement Rules of Thumb

Higher level of sophistication is better, in general.

- Having more information about the possible values/attributes of a variable is better than having less
 - Continuous level variables are preferred over ordinal-level ones
 - Ordinal level variables are preferred over nominal-level
 - We prefer ordinal and nominal level variables with more, rather than fewer, categories

Why use statistics?

- Generally, in the social sciences, we want to ask and answer questions about what's going on in the population. We want to investigate how some construct operates in the population.
- Oftentimes, however, we don't have the ability, time, resources, or energy to collect information from every case in the population.
- Instead we rely on data from a small subset of the population (AKA a sample). And if the sample is representative of the population, we can use the findings from our sample to make inferences/generalizable claims about how a construct operates in the population.

Important Statistics Terms

Population:

- The total collection of cases or observations that a researcher is interested in. For example, if a researcher is interested in studying Californian police officers, the population would be "all police officers in California."

N:

- A statistical symbol representing the total number of cases in a population. Population size.

Parameters:

- Statistical calculations used to describe what is average or typical of a population. For example: mean, median, mode, standard deviation, variance, range, etc.

Sample:

- A subset of (i.e. a smaller number of cases than) the population.

n

- A statistical symbol representing the total number of cases in a sample. Sample size.

Statistics:

- Statistical calculations used to describe what is average or typical of a sample. For example: mean, median, mode, standard deviation, variance, range, etc. Just like parameters, but this time calculated for the sample.

Unit of Analysis

The entity being studied. The level of social life being investigated. What each case/observation represents.

Some examples of investigations at different units include comparing:

- Individuals
 - People, students, adults, children
- Social groups
 - Families, racial groups
- Organizational units
 - Departments in an organization
- Organizations/Institutions
 - Universities, Companies, Churches
- Artifacts
 - Books, photos, newspaper articles, videos, websites
- Geographical units
 - Countries, states, counties, cities, towns, Census tracts, Census blocks

Dependent and Independent Variables

Dependent Variable:

- The variable to be explained. The main focus of your research.
- The outcome
- The effect
- The Y variable

Independent Variable:

- The variable expected to explain/influence the dependent variable
- The predictor
- The presumed "cause"
- The X variable

Causation in the Social Sciences

Difference between a causal and an associational relationship between the independent and dependent variables.

In order to say that there is a causal relationship between our independent and dependent variable (i.e. that the values of the independent variable “caused” changes in the value of our dependent variable), we must meet 3 criteria:

1. Time order: The independent variable must precede (come before) the dependent variable
2. Association: There must be evidence of a statistical association between the independent and dependent variable
3. Absence of spuriousness: the relationship between the two variables cannot be explained by some unaccounted for third variable.